



Customer Shopping Behavior Analysis

Exploring purchasing trends, demographics, and product performance across 3,900 retail transactions

Tools & Dataset



Tech Stack

Python

PostgreSQL

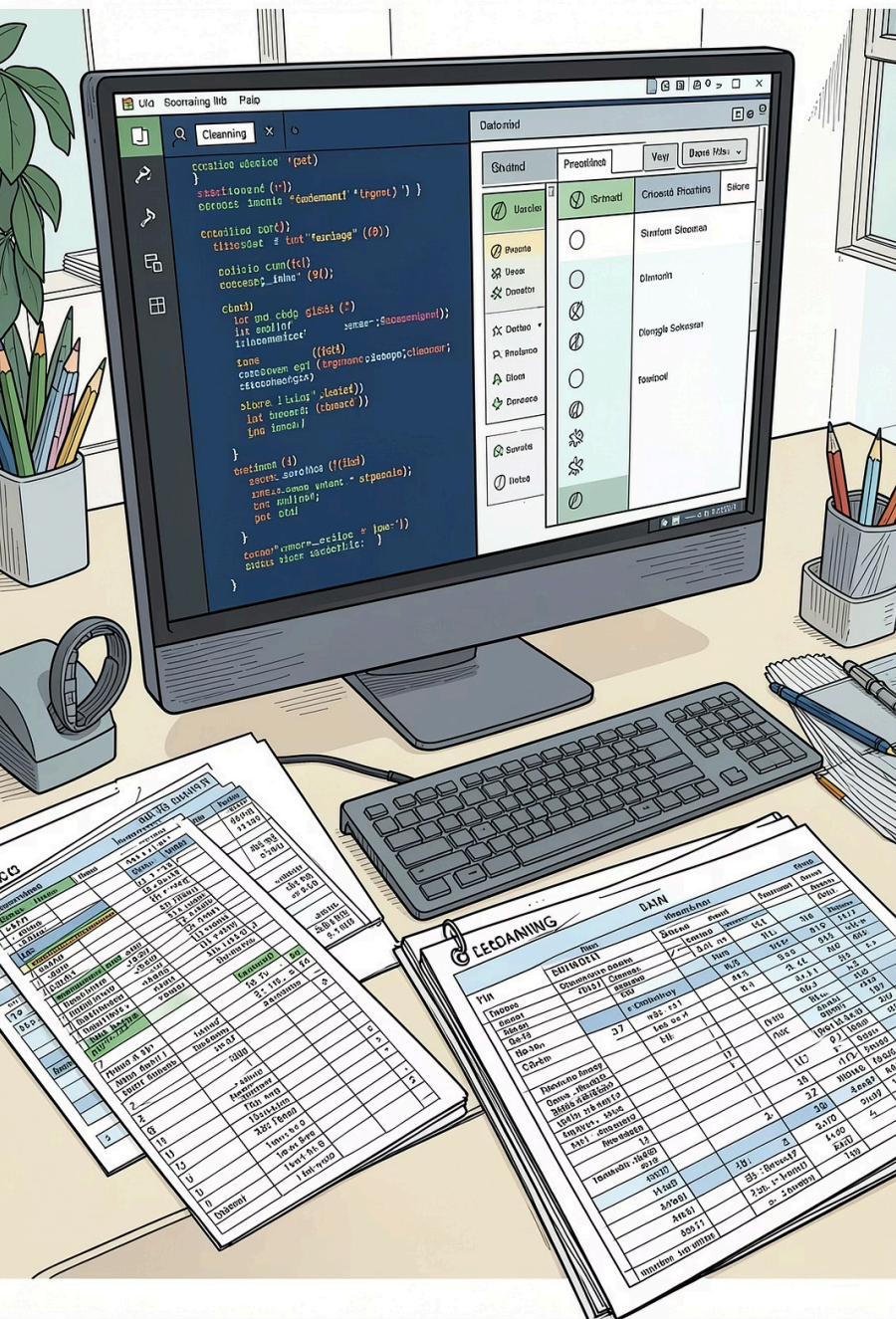
Tableau

Git

Dataset

3,900 records · 18 features

Customer info, purchases,
subscriptions, reviews



Data Cleaning & Preparation

1

Clean

Fill missing values, rename columns, remove redundancy

2

Engineer

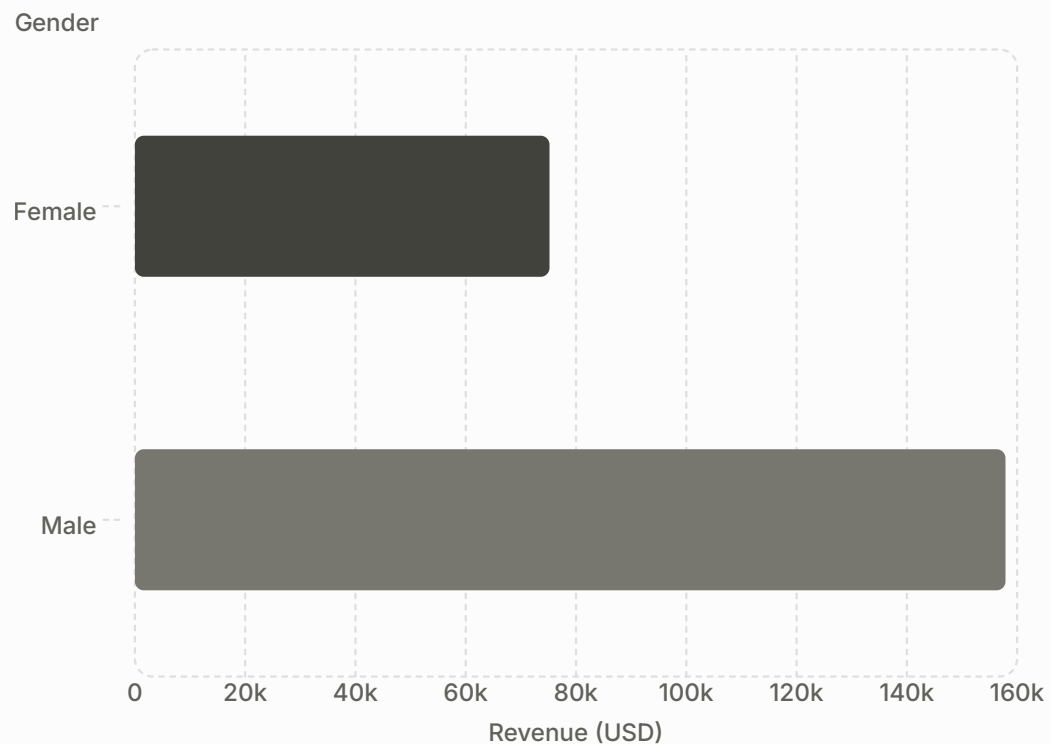
Age groups, purchase frequency, standardized categories

3

Validate

Check duplicates, verify consistency, prepare for SQL

Revenue by Gender

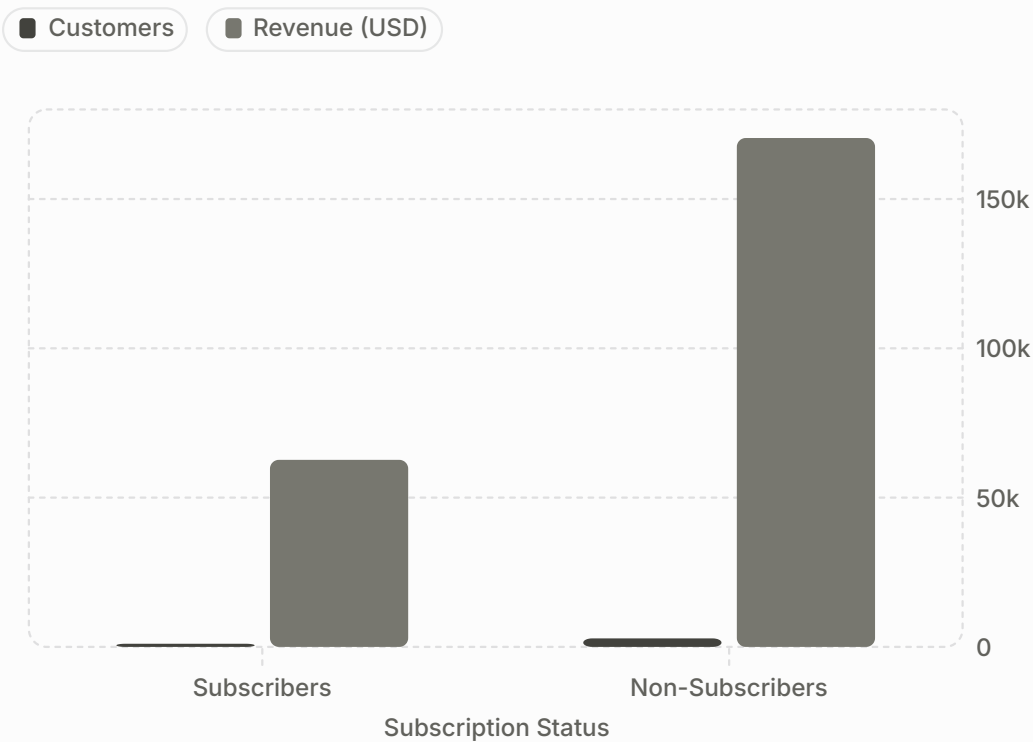


Key Finding

Male customers generated **2.1x more revenue** than female customers

📌 Male: \$157,890 vs Female: \$75,191

Subscribers vs. Non-Subscribers



Insight

73% of customers are non-subscribers, yet average spend is nearly identical

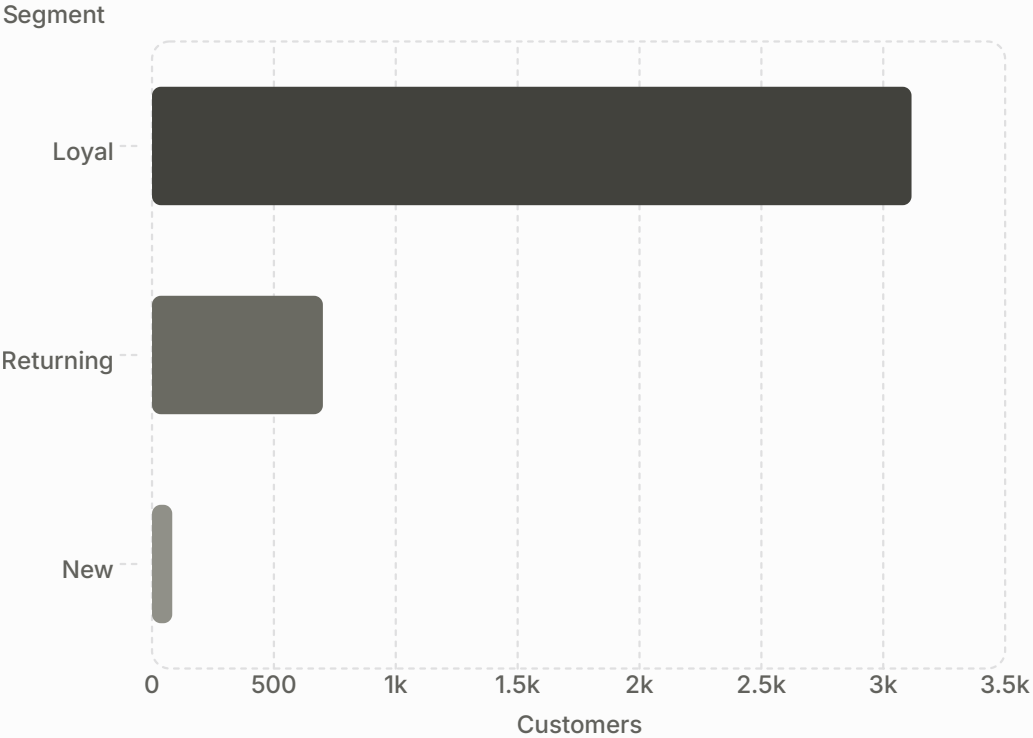
Avg Spend (Yes)

\$59.49

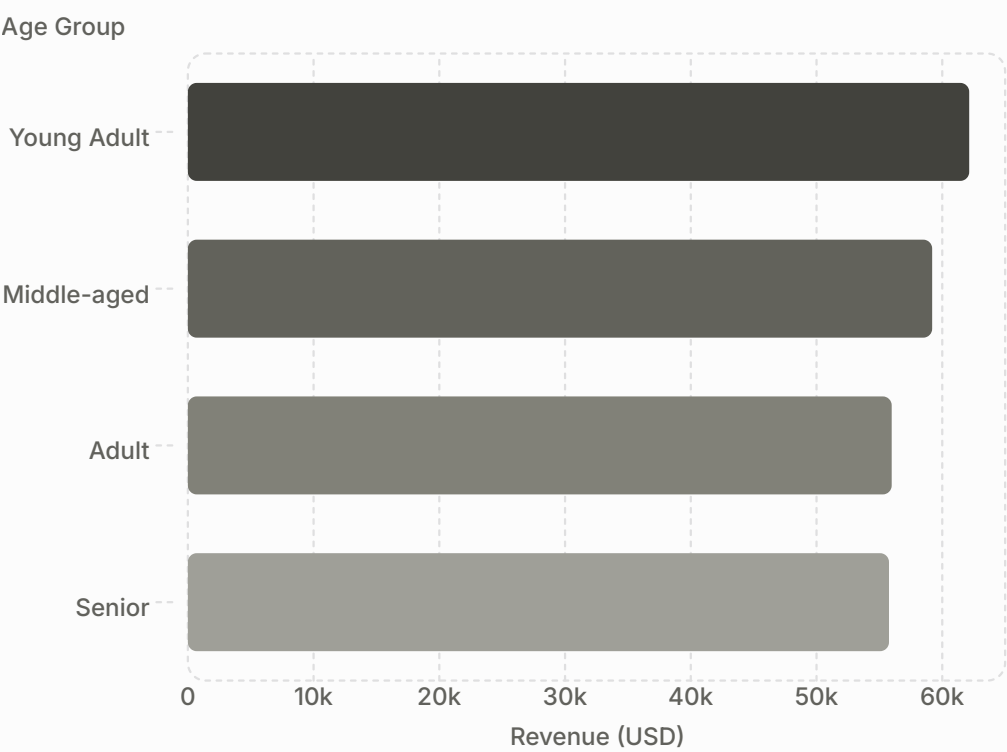
Avg Spend (No)

\$59.87

Customer Segmentation & Revenue by Age



Revenue by Age Group



Top Products by Rating & Discount Dependency

Top 5 Rated Products

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discounted Products

Product	Discount Rate
Hat	50%
Sneakers	49%
Coat	49%
Sweater	48%
Pants	47%

Tableau Dashboard



Interactive dashboard with filters for subscription status, gender, category, and shipping type — enabling real-time exploration of customer behavior

Key Insights

Clothing Dominates

Highest revenue across all product categories

Young Adults Lead

Highest total revenue contribution by age group

Subscription Gap

73% non-subscribers with consistent average spend

Loyal Base

3,116 loyal customers — strong repeat purchase potential



Business Recommendations



Boost Subscriptions

Exclusive member benefits to convert non-subscribers



Invest in Clothing

Allocate more marketing budget to top-revenue category



Target Young Adults

Campaigns focused on highest-revenue age group



Loyalty Rewards

Incentivize repeat buyers and optimize shipping options